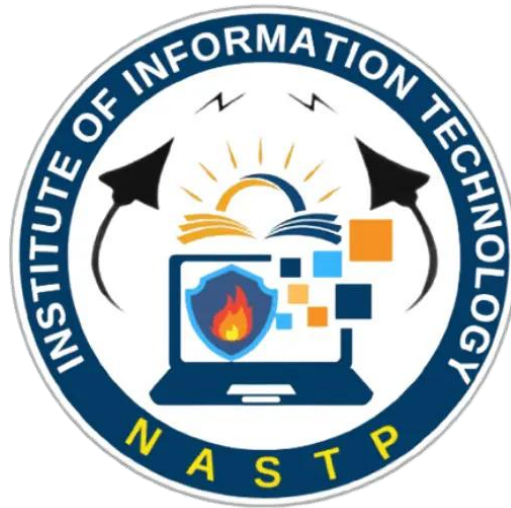




NASTP Institute of Information Technology
Department of Computer Science

Semester Project Report

“Anemia Classification and Prediction Using Ensemble Machine Learning”

Field	Details
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Semester	IV
Section	BSCS Fall 2024
Subject	Machine Learning (AI233)

1. Executive Summary

This report presents a comprehensive analysis of an anemia prediction system developed using machine learning techniques on a large-scale clinical dataset. The system leverages hematological blood parameters to classify patients as anemic or non-anemic, providing a decision-support tool for clinical settings.

The project encompasses exploratory data analysis (EDA), feature engineering, model training, and a real-time interactive prediction application. Two ensemble learning algorithms — Random Forest and XGBoost — were trained and evaluated, with the Random Forest classifier emerging as the best-performing model.

Dataset Size	Best Model	Accuracy	AUC Score
153,728 records	Random Forest	93.48%	~0.98

Key findings from this project include identification of Hemoglobin, Hematocrit, and RBC count as the most predictive features for anemia detection, and an ensemble voting approach that combines both models for robust predictions.

2. Introduction

2.1 Background

Anemia is a global health condition characterized by a deficiency of red blood cells or hemoglobin in the blood, resulting in reduced oxygen-carrying capacity. According to the World Health Organization (WHO), anemia affects approximately 1.62 billion people worldwide, representing nearly 24.8% of the global population. It is particularly prevalent among children under five, pregnant women, and elderly individuals.

Early and accurate detection of anemia is critical for timely clinical intervention. Traditional diagnosis relies on manual interpretation of complete blood count (CBC) panels, which can be time-consuming and prone to variability. Machine learning-based prediction systems offer an opportunity to automate and standardize this process, enabling faster and more consistent clinical decisions.

2.2 Objectives

- Perform exploratory data analysis on a large-scale hematological dataset to understand feature distributions and class imbalances.
- Identify key blood parameters most predictive of anemia using correlation analysis and feature importance metrics.
- Develop and compare machine learning models (Random Forest and XGBoost) for binary anemia classification.

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- Build an interactive web-based application using Streamlit to enable real-time prediction from user-supplied blood parameter values.
- Evaluate model performance using standard classification metrics including accuracy, precision, recall, F1-score, and AUC-ROC.

2.3 Scope

The project focuses exclusively on binary classification (anemia vs. no anemia) based on 14 hematological features. The analysis does not distinguish between anemia subtypes (e.g., iron-deficiency, B12 deficiency, hemolytic) but provides a general screening tool. The dataset used is ANERBC_CORELAB_152K, a real-world clinical dataset containing 153,728 patient records.

3. Dataset Description

3.1 Overview

The dataset used in this project is a large-scale clinical laboratory dataset sourced from a core hematology lab. It contains complete blood count and biochemical markers for 153,728 patients, recorded between 2024 and the project date.

Property	Details
File Name	ANERBC_CORELAB_152K_20240603_R01.csv
Total Records	153,728
Total Features	18 (17 input + 1 target)
Missing Values	None (complete dataset)
Target Variable	Result (0 = No Anemia, 1 = Anemia)
Class Distribution	No Anemia: 100,395 (65.3%) Anemia: 53,333 (34.7%)
Data Types	15 float64, 3 int64

	Age	Gender	Hemoglobin(g/dL)	Hematocrit(fL)	RBC(mil/uL)	MCV(fL)	MCH(pg)	MCHC(g/dL)	RDW(fL)	Serum_Iron(ug/dL)	Ferritin(ng/mL)	TIBC(ug/dL)	Vitamin_B12(pg/mL)	Folate(n
0	43	1	17.22	31.75860	6.15	76.54	28.15	33.81	9.881314	91.10	46.31	288.19	444.09	
1	23	0	10.45	10.50282	3.33	86.62	33.39	32.81	9.900666	68.68	113.89	326.61	564.58	
2	73	1	11.94	14.77844	4.12	94.14	29.69	34.04	12.643002	59.59	49.61	336.92	462.40	
3	32	0	14.92	23.87705	5.35	80.23	23.16	32.76	10.189210	73.98	400.00	292.48	355.74	
4	46	0	11.04	12.15324	3.96	84.61	30.07	36.94	12.649195	61.06	5.00	306.00	564.46	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
153723	54	1	9.90	10.24445	3.31	83.35	30.43	31.74	12.185770	120.11	58.56	346.25	582.07	
153724	35	0	11.85	13.90158	3.74	95.42	27.11	31.50	12.328264	20.00	108.15	353.69	443.81	
153725	63	1	15.21	25.52394	5.59	81.07	26.89	33.80	10.936343	97.58	109.29	310.10	397.95	
153726	1	1	14.11	22.30613	4.91	86.85	30.67	31.38	10.109340	127.54	192.23	285.69	427.60	
153727	31	1	14.84	21.78484	4.84	77.51	28.56	33.24	11.052926	20.00	280.21	291.97	295.27	

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3.2 Feature Description

The dataset contains 17 independent features consisting of demographic information and hematological blood parameters routinely measured in clinical laboratory settings:

Feature	Unit	Normal Range	Clinical Significance
Age	Years	1 – 90	Demographic variable
Gender	0/1 (F/M)	—	Demographic variable
Hemoglobin	g/dL	12.0 – 18.0	Primary anemia indicator
Hematocrit	fL	36.0 – 50.0	RBC volume fraction
RBC Count	mil/uL	4.0 – 6.0	Red blood cell count
MCV	fL	80.0 – 100.0	Mean corpuscular volume
MCH	pg	27.0 – 33.0	Mean corpuscular hemoglobin
MCHC	g/dL	32.0 – 36.0	Mean corpuscular Hb concentration
RDW	fL	11.5 – 14.5	Red cell distribution width
Serum Iron	ug/dL	60.0 – 170.0	Iron level in blood
Ferritin	ng/mL	30.0 – 300.0	Iron storage protein
TIBC	ug/dL	250.0 – 400.0	Total iron-binding capacity
Vitamin B12	pg/mL	200.0 – 900.0	Cobalamin level
Folate	ng/mL	2.7 – 17.0	Folic acid level
CRP	mg/L	0.0 – 5.0	C-reactive protein (inflammation)
WBC	10 ³ /uL	4.5 – 11.0	White blood cell count
Platelets	10 ³ /uL	150.0 – 400.0	Platelet count

3.3 Descriptive Statistics

Key statistical summaries derived from `data.describe()` provide insight into feature distributions and value ranges:

Feature	Mean	Std Dev	Min	Max	Median
Age (years)	34.74	17.49	1	90	35.00

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Feature	Mean	Std Dev	Min	Max	Median
Hemoglobin (g/dL)	13.08	2.12	6.00	18.00	13.25
Hematocrit (fL)	17.99	5.67	3.60	39.27	17.92
RBC (mil/uL)	4.47	0.78	2.00	7.16	4.52
Serum Iron (ug/dL)	83.92	31.11	20.00	200.00	83.97
Ferritin (ng/mL)	118.50	69.34	5.00	400.00	116.87
TIBC (ug/dL)	311.48	44.69	200.00	450.00	310.13

Data columns (total 18 columns):

#	Column	Non-Null Count	Dtype
0	Age	153728 non-null	int64
1	Gender	153728 non-null	int64
2	Hemoglobin(g/dL)	153728 non-null	float64
3	Hematocrit(fL)	153728 non-null	float64
4	RBC(mil/uL)	153728 non-null	float64
5	MCV(fL)	153728 non-null	float64
6	MCH(pg)	153728 non-null	float64
7	MCHC(g/dL)	153728 non-null	float64
8	RDW(fL)	153728 non-null	float64
9	Serum_Iron(ug/dL)	153728 non-null	float64
10	Ferritin(ng/mL)	153728 non-null	float64
11	TIBC(ug/dL)	153728 non-null	float64
12	Vitamin_B12(pg/mL)	153728 non-null	float64
13	Folate(ng/mL)	153728 non-null	float64
14	CRP(mg/L)	153728 non-null	float64
15	WBC(10*3/uL)	153728 non-null	float64
16	Platelets(10*3/uL)	153728 non-null	float64
17	Result	153728 non-null	int64

4. Exploratory Data Analysis

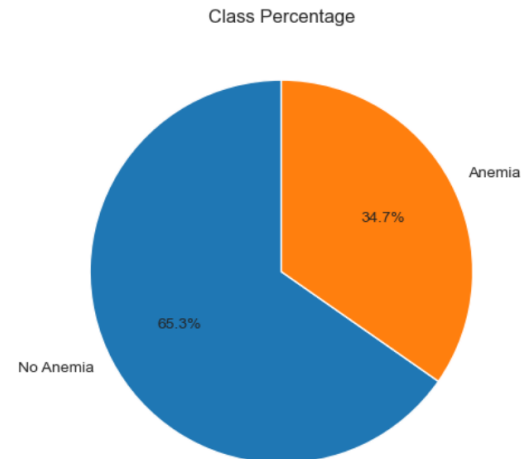
4.1 Target Variable Distribution

The dataset exhibits a moderate class imbalance between the two outcome categories:

- No Anemia (Class 0): 100,395 records — 65.3% of the dataset
- Anemia (Class 1): 53,333 records — 34.7% of the dataset

The approximately 2:1 ratio of non-anemic to anemic cases reflects realistic clinical prevalence. While not severely imbalanced, this distribution was considered during model evaluation, particularly in examining recall for the minority (anemic) class.

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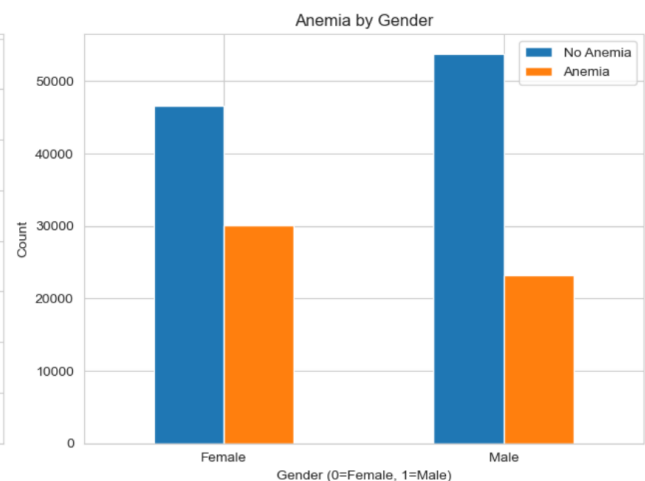


4.2 Gender Analysis

The dataset is nearly gender-balanced with 77,001 males (50.1%) and 76,727 females (49.9%). Anemia rates differ meaningfully by gender:

Gender	No Anemia Rate	Anemia Rate
Female (0)	60.73%	39.27%
Male (1)	69.87%	30.13%

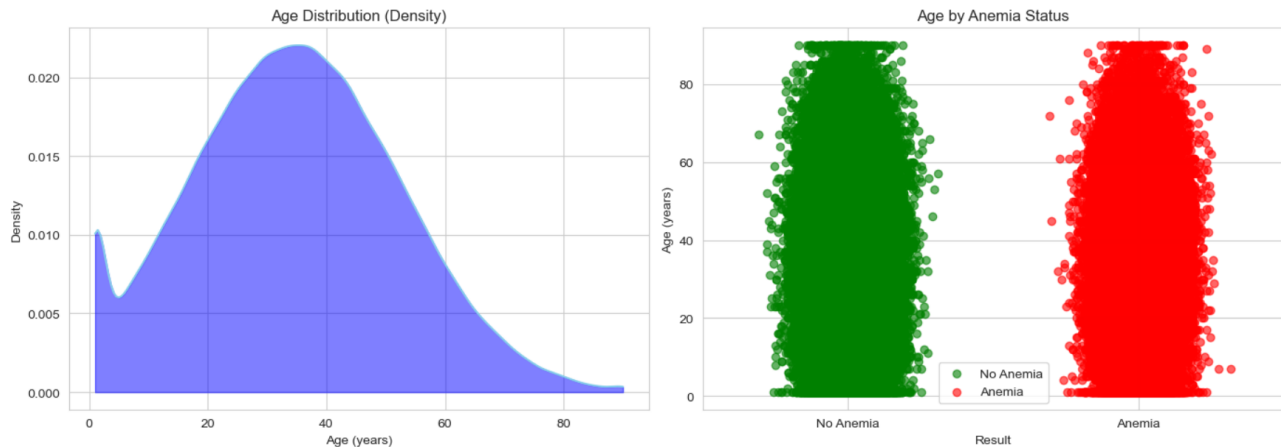
Females exhibit a notably higher anemia prevalence (39.27% vs. 30.13%), consistent with clinical literature attributing this to menstrual blood loss and pregnancy-related iron demands.



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4.3 Age Distribution

Patient ages range from 1 to 90 years with a mean of 34.74 years and a median of 35 years, indicating a roughly symmetric distribution skewed slightly toward younger-to-middle-aged patients. The scatter plot analysis of age by anemia status shows no strong age-based differentiation, with both anemic and non-anemic patients distributed broadly across all age groups.



4.4 Feature Correlations with Anemia

Pearson correlation analysis with the target variable (Result) revealed the following key relationships:

Feature	Correlation	Interpretation
Hemoglobin (g/dL)	-0.775	Strongest negative predictor
Hematocrit (fL)	-0.728	Strong negative correlation
RBC (mil/uL)	-0.692	Strong negative correlation
Serum Iron (ug/dL)	-0.269	Moderate negative correlation
Ferritin (ng/mL)	-0.254	Moderate negative correlation
TIBC (ug/dL)	+0.347	Strongest positive predictor
RDW (fL)	+0.248	Moderate positive correlation
Gender	-0.096	Weak negative correlation

These correlations align well with established clinical knowledge: lower hemoglobin, hematocrit, and RBC values are hallmark indicators of anemia, while elevated TIBC reflects compensatory iron transport increase in iron-deficient states.

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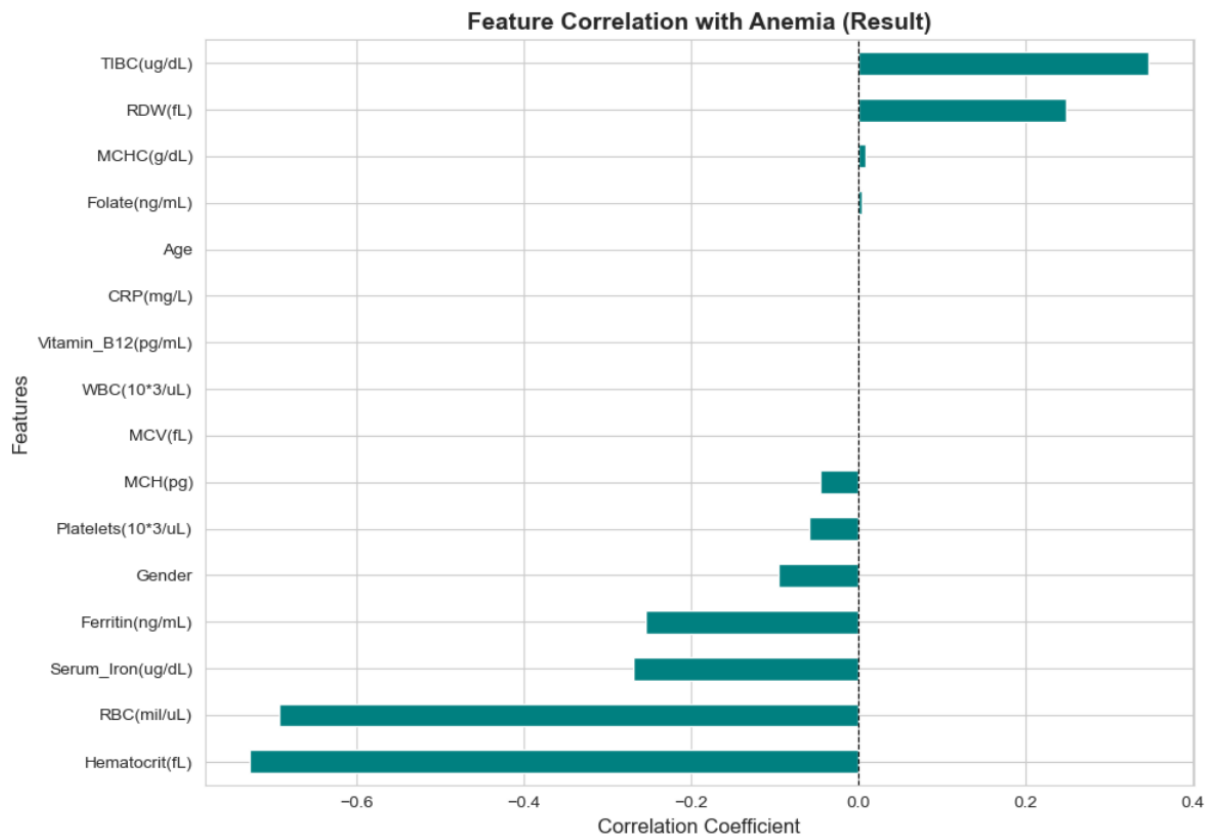
Correlation Values:

Correlation with Target (Result):

Result	1.000000
TIBC(ug/dL)	0.347202
RDW(fL)	0.248246
MCHC(g/dL)	0.008947
Folate(ng/mL)	0.004848
Age	-0.000243
CRP(mg/L)	-0.000280
Vitamin_B12(pg/mL)	-0.001078
WBC(10 ³ /uL)	-0.001577
MCV(fL)	-0.002427
MCH(pg)	-0.045734
Platelets(10 ³ /uL)	-0.059365
Gender	-0.095965
Ferritin(ng/mL)	-0.254056
Serum_Iron(ug/dL)	-0.269098
RBC(mil/uL)	-0.692084
Hematocrit(fL)	-0.727824
Hemoglobin(g/dL)	-0.774613

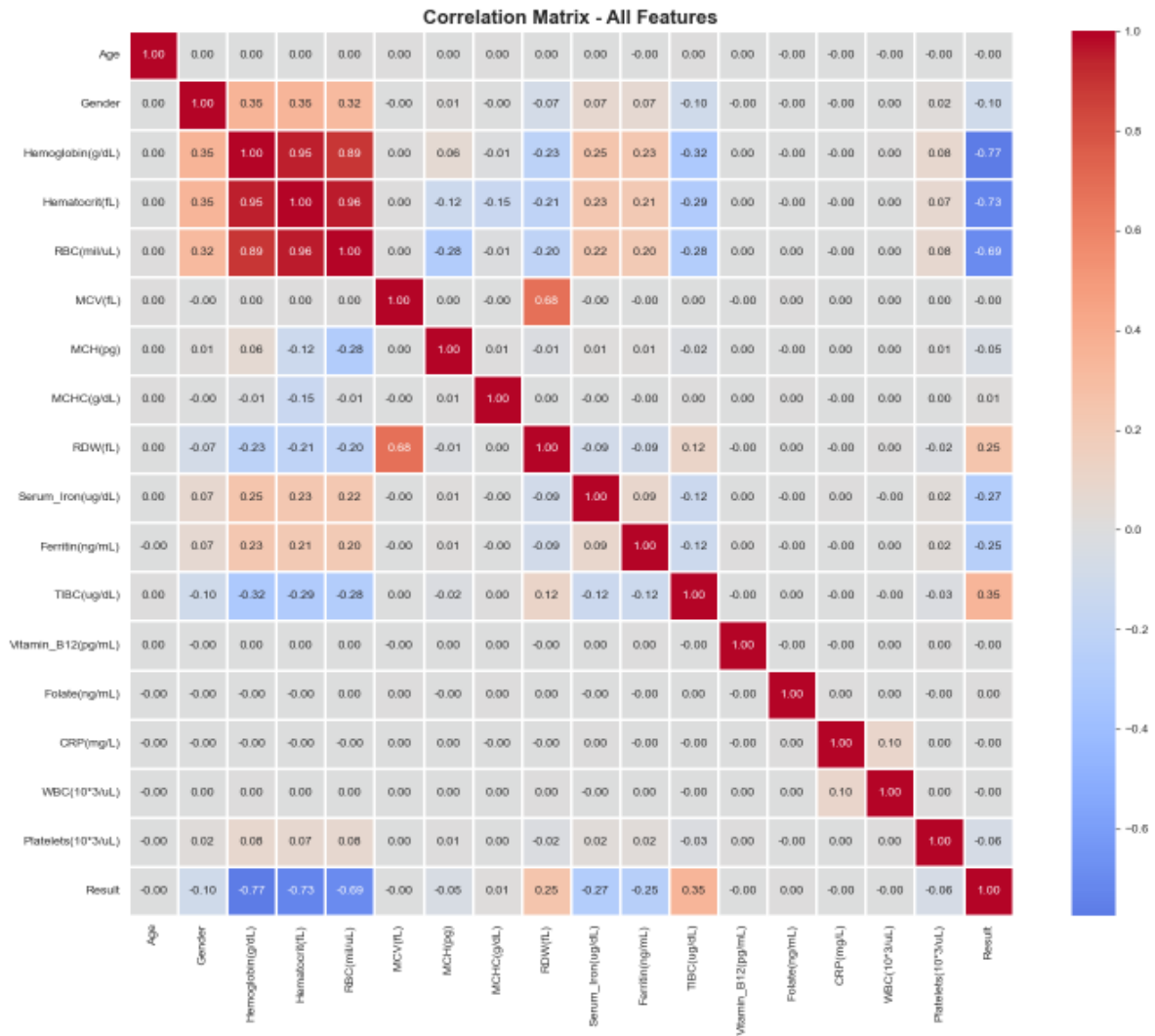
Name: Result, dtype: float64

Coefficient Variation:



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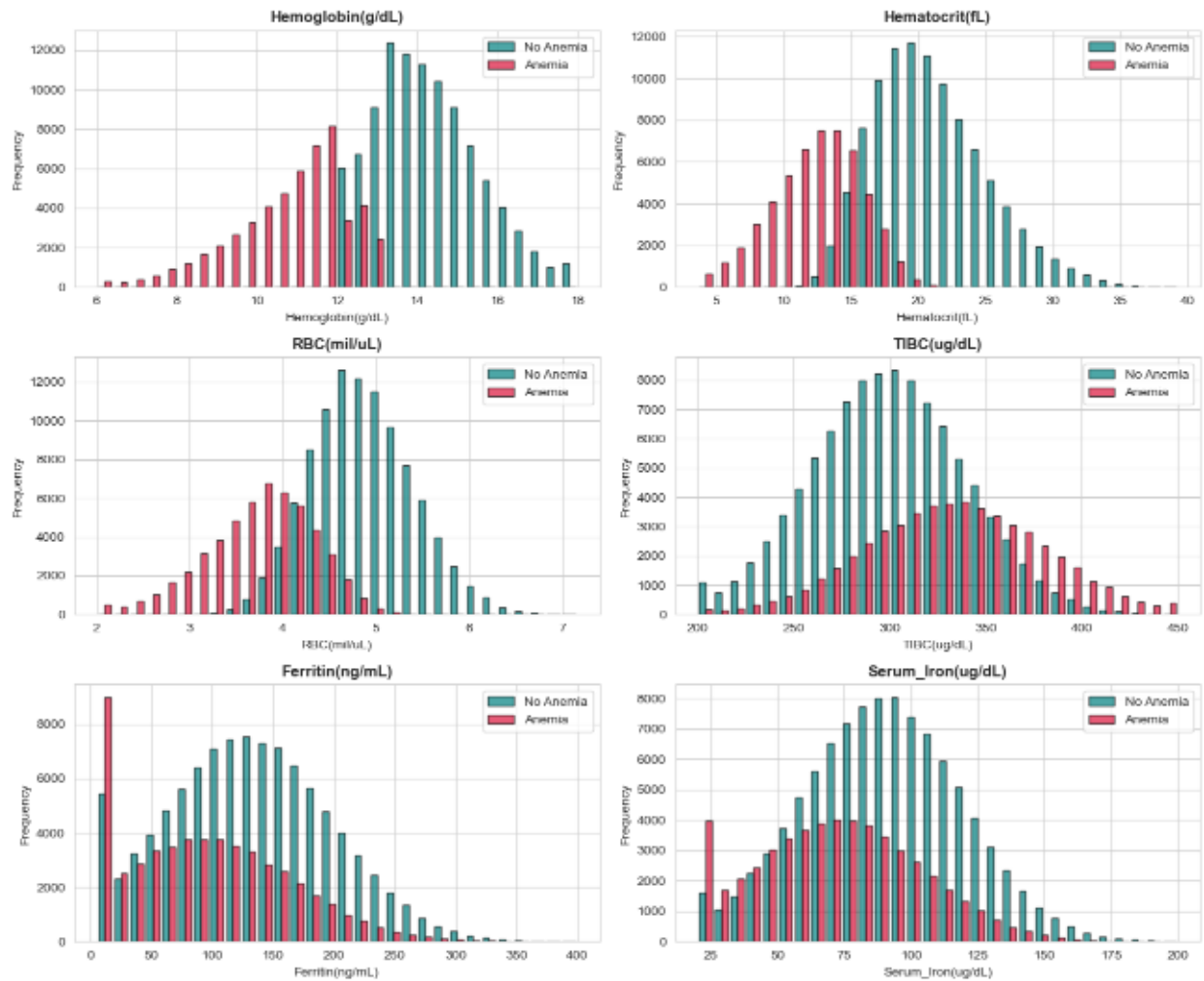
Correlation Matrix:



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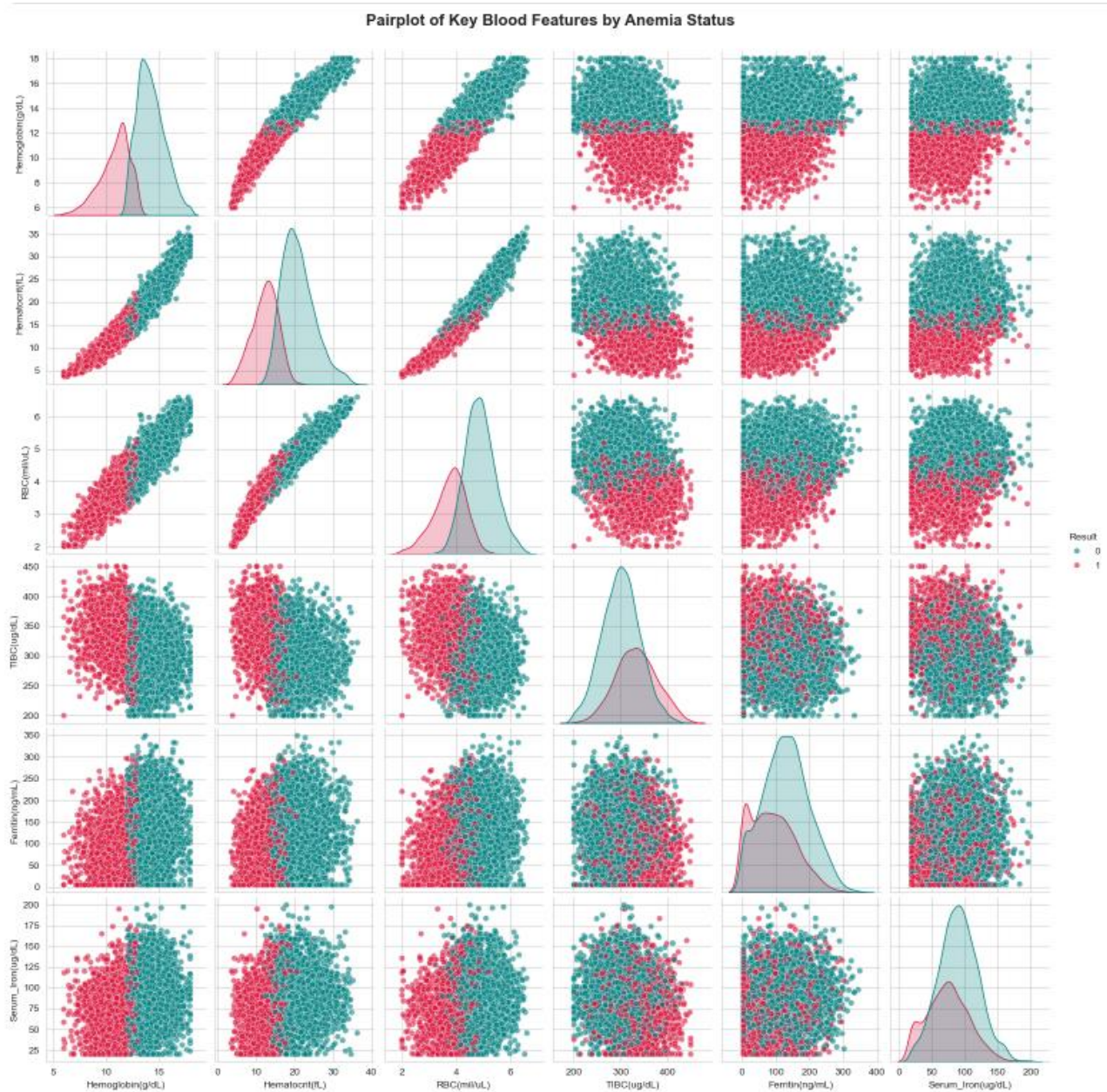
Blood Parameters Varying Anemia:

Blood Parameters by Anemia Status



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Pairplot for Selected Parameters:



4.5 Outlier Analysis

IQR-based outlier detection was performed on all numeric features. The following features exhibited the highest outlier proportions:

Feature	Outlier Count	Percentage
CRP (mg/L)	7,233	4.71%
MCH (pg)	1,610	1.05%

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Feature	Outlier Count	Percentage
Hemoglobin (g/dL)	1,508	0.98%
RBC (mil/uL)	1,377	0.90%
RDW (fL)	1,275	0.83%

CRP showed the highest outlier rate (4.71%), likely reflecting patients with acute inflammatory conditions. The overall outlier proportion across features was low (under 5%), suggesting the dataset is relatively clean. No outlier removal was performed, as these values may represent clinically valid extreme cases.

	Count	Percentage
CRP(mg/L)	7233	4.705063
MCH(pg)	1610	1.047304
Hemoglobin(g/dL)	1508	0.980953
RBC(mil/uL)	1377	0.895738
RDW(fL)	1275	0.829387
MCV(fL)	1037	0.674568
TIBC(ug/dL)	843	0.548371
WBC(10*3/uL)	604	0.392902
Hematocrit(fL)	543	0.353221
Platelets(10*3/uL)	494	0.321347

5. Feature Selection & Preprocessing

5.1 Selected Features

Based on clinical relevance and correlation analysis, 14 blood parameters were selected as model input features, excluding Age, Gender, and CRP which showed negligible linear correlations with the target:

Selected Features (14 Total)	Rationale
Hemoglobin, Hematocrit, RBC	Primary CBC anemia markers (highest correlation)
Serum Iron, Ferritin, TIBC	Iron metabolism indicators
MCV, MCH, MCHC, RDW	Red cell morphology parameters
WBC, Platelets	Broader blood panel context
Vitamin B12, Folate	Nutritional deficiency markers

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5.2 Data Splitting

The dataset was partitioned using a stratified train-test split to preserve the class distribution across both subsets:

- Training Set: 80% — 122,982 records
- Test Set: 20% — 30,746 records
- Random State: 42 (for reproducibility)
- Stratify: Yes (maintains anemia/non-anemia ratio)

5.3 Feature Scaling

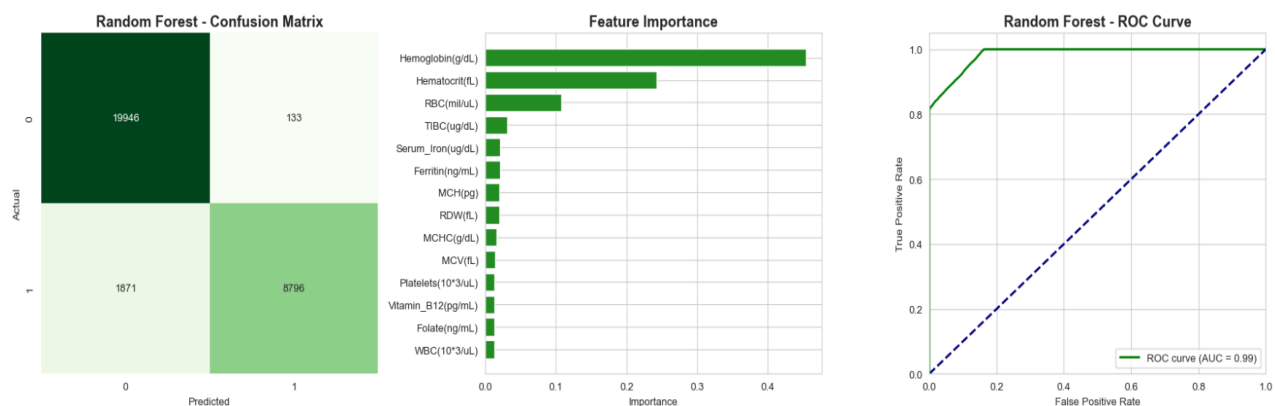
StandardScaler from scikit-learn was applied to normalize all 14 features to zero mean and unit variance. The scaler was fit exclusively on the training set and applied to both training and test sets to prevent data leakage. The fitted scaler was serialized as scaler.pkl for reuse in the prediction application.

6. Model Implementation

6.1 Random Forest Classifier

Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the mode of their individual predictions. It is robust to overfitting due to bootstrap aggregation (bagging) and random feature selection at each split.

Hyperparameter	Value
n_estimators	100
random_state	42
n_jobs	-1 (all cores)
max_features	sqrt (default)
Training Time	7.08 seconds



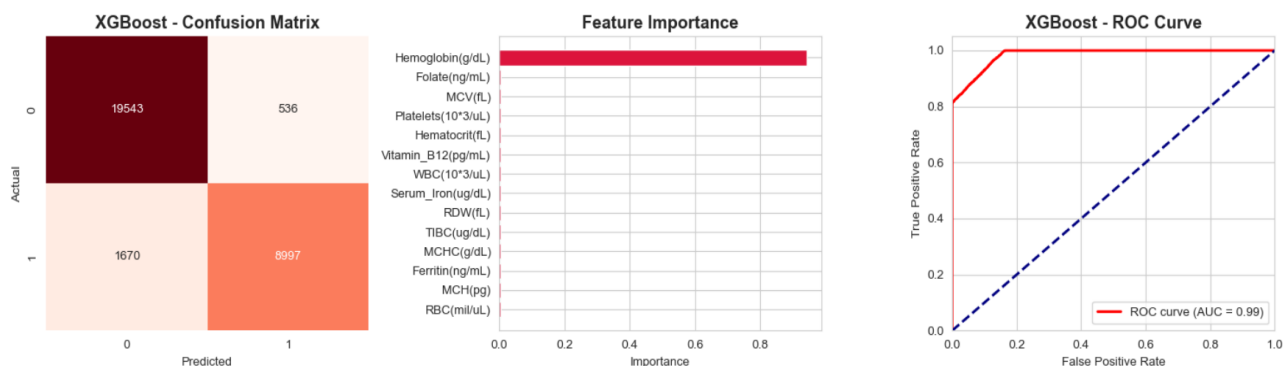
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6.2 XGBoost Classifier

XGBoost (Extreme Gradient Boosting) is a scalable gradient boosting framework that builds trees sequentially, with each tree correcting errors of the previous. It is known for high performance and computational efficiency.

Hyperparameter	Value
random_state	42
eval_metric	logloss
n_estimators	100 (default)
learning_rate	0.3 (default)
Training Time	1.02 seconds

XGBoost trained approximately 7x faster than Random Forest (1.02s vs. 7.08s), demonstrating its computational efficiency advantage on large datasets.



6.3 Ensemble Prediction Strategy

The Streamlit application implements a simple majority voting ensemble: if at least one of the two models predicts anemia, the ensemble result is flagged as anemia. The confidence score is calculated as the average of both models' maximum class probability. This conservative approach minimizes false negatives, which is clinically preferable — it is safer to over-refer than to miss a true anemia case.

7. Model Evaluation & Results

7.1 Performance Metrics

Both models were evaluated on the held-out test set (30,746 records) using four standard classification metrics:

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Model	Accuracy	Precision	Recall	F1-Score
Random Forest	93.48%	98.51%	82.46%	89.77%
XGBoost	92.83%	94.38%	84.34%	89.08%

Random Forest outperformed XGBoost on Accuracy, Precision, and F1-Score, making it the best overall model for this task. XGBoost achieved a marginally higher recall (84.34% vs. 82.46%), indicating it correctly identified slightly more true anemia cases at the cost of more false positives.

	Accuracy	Precision	Recall	F1-Score	Training Time
Random Forest	0.934821	0.985105	0.824599	0.897734	7.076544
XGBoost	0.928251	0.943774	0.843442	0.890792	1.020638

BEST MODEL: Random Forest
F1-Score: 0.8977
Accuracy: 0.9348

7.2 Metric Interpretation

Metric	Random Forest	XGBoost
Accuracy	93.48% — correctly classifies 28,748 of 30,746 cases	92.83% — correctly classifies 28,548 of 30,746 cases
Precision	98.51% — when RF predicts anemia, it is correct 98.5% of the time	94.38% — slightly more false positives than RF
Recall	82.46% — detects 82.5% of all actual anemia cases	84.34% — detects slightly more anemia cases
F1-Score	89.77% — best harmonic balance of precision and recall	89.08% — slightly lower balance

7.3 Feature Importance

Random Forest's built-in feature importance (mean decrease in impurity) highlighted the following top predictors:

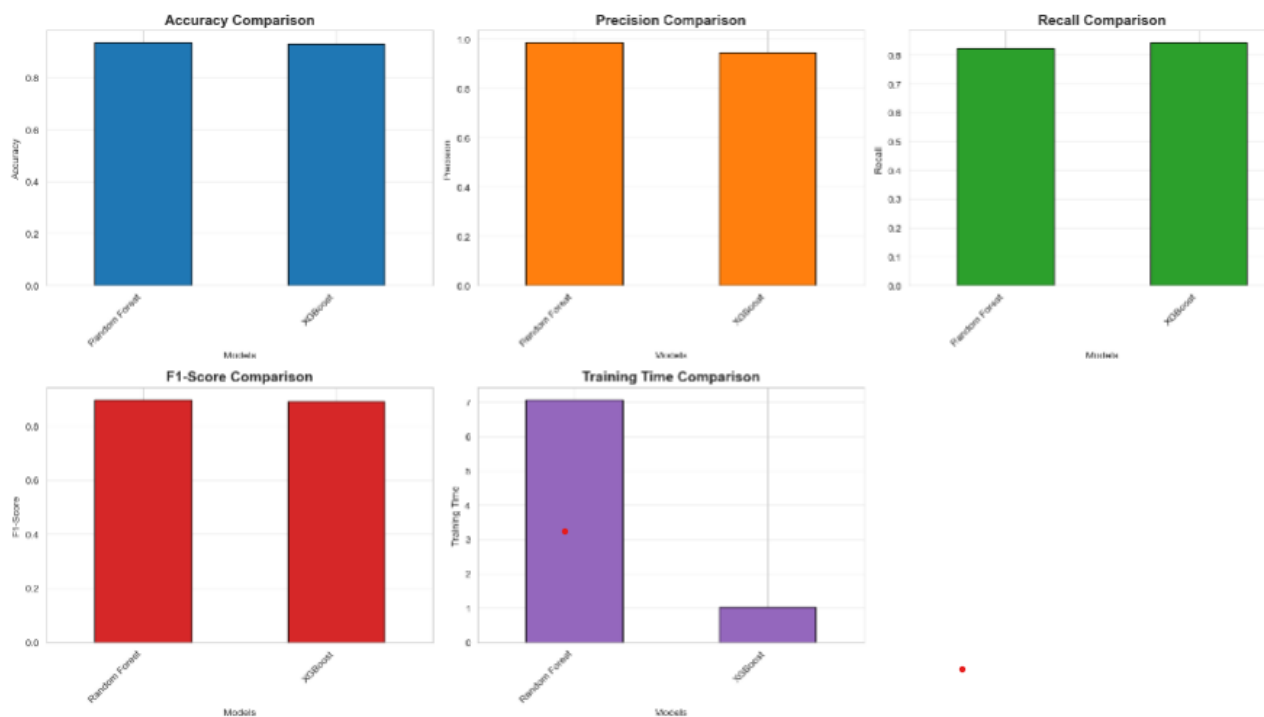
- Hemoglobin (g/dL) — consistently the highest-importance feature, reflecting its direct role in anemia diagnosis.
- Hematocrit (fL) — second most important, closely related to hemoglobin.
- RBC Count (mil/uL) — third most important, directly measuring red cell abundance.
- TIBC (ug/dL) — fourth, reflecting iron transport capacity elevation in deficiency states.
- Serum Iron and Ferritin — mid-range importance, contributing supplementary diagnostic signals.

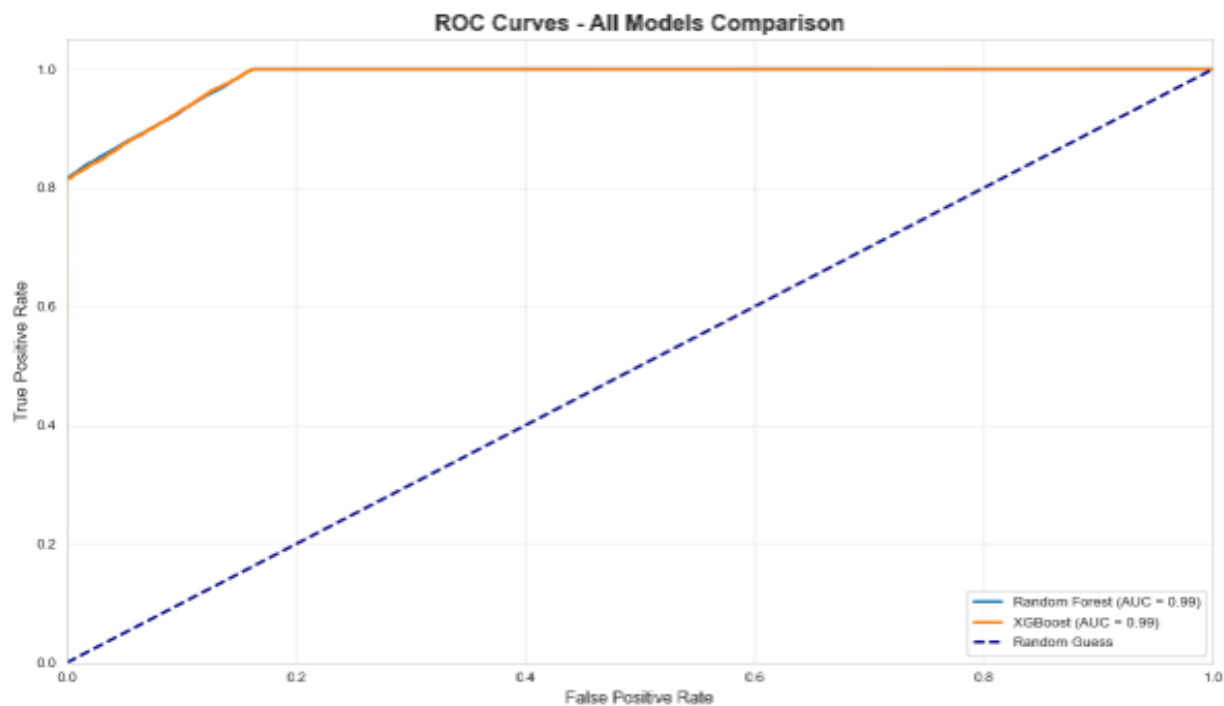
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Features such as WBC, Vitamin B12, CRP, and Folate showed lower importance scores, suggesting limited discriminative power for binary anemia classification in this dataset. This is consistent with these markers being more relevant for specific anemia subtypes.

7.4 Best Model Conclusion

Random Forest is selected as the primary production model due to its superior precision (98.51%) and F1-Score (89.77%). Its extremely high precision is particularly valuable in a clinical support context — when the model raises an anemia flag, there is very high confidence in the prediction. The ensemble approach in the application combines both models to leverage XGBoost's marginally better recall as a safety net.





8. Streamlit Web Application

8.1 Overview

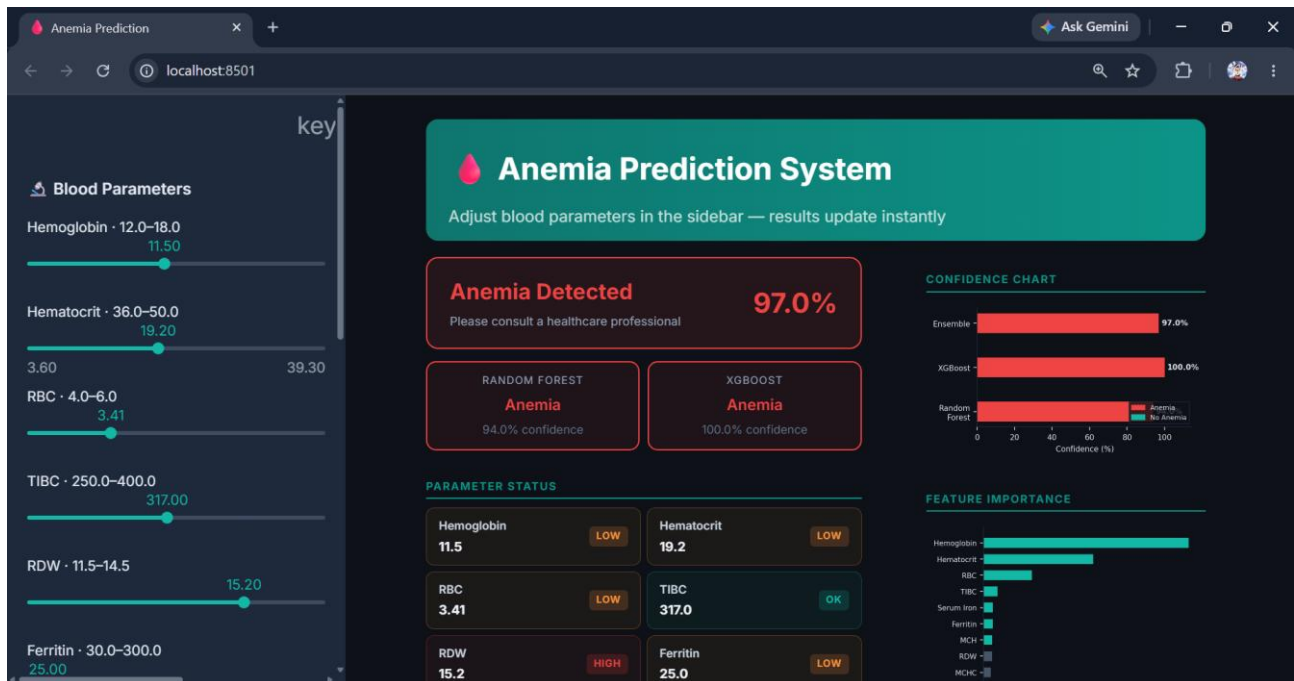
A production-grade interactive prediction application was built using Streamlit. The app allows clinicians or analysts to input patient blood parameters via intuitive sliders and receive instant anemia predictions from both models along with an ensemble decision.

8.2 Application Architecture

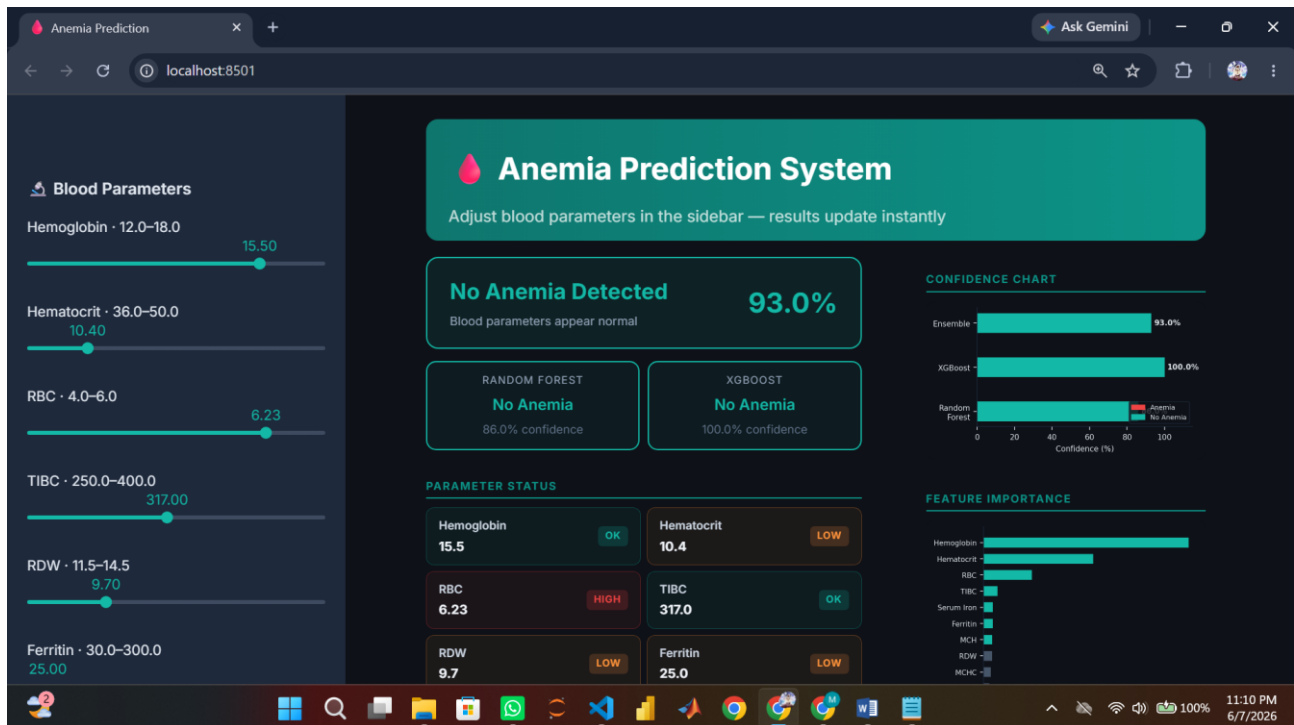
- **Frontend:** Streamlit with custom CSS styling (Inter font, dark-mode palette, teal/red color scheme)
- **Backend:** Pre-trained Random Forest and XGBoost models loaded from serialized .pkl files via joblib
- **Preprocessing:** The same StandardScaler used during training is loaded and applied to user input
- **Prediction:** Both models predict independently; ensemble vote determines the final result
- **Visualization:** Matplotlib charts embedded for real-time confidence and feature importance display

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Anemia Detected:



No Anemia Detected:



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8.3 Key Features

Feature	Description
Real-time Prediction	Results update instantly as sliders are adjusted — no submit button required
Parameter Status Grid	Each of 14 blood parameters displays a color-coded badge (OK / Low / High) based on clinical reference ranges
Dual Model Display	Individual predictions from Random Forest and XGBoost shown side-by-side with confidence %
Ensemble Decision Banner	A prominent result banner (red for Anemia, teal for No Anemia) with overall confidence score
Confidence Chart	Horizontal bar chart showing prediction confidence for all three models (RF, XGBoost, Ensemble)
Feature Importance Chart	Bar chart of RF feature importances, color-coded by above/below median importance
Healthy Range Display	Slider labels show normal reference ranges for each parameter

8.4 Model Loading & Serialization

Models are cached using `@st.cache_resource` to avoid reloading on every user interaction. Three files are required at runtime:

- `rf_model.pkl` — Trained Random Forest classifier
- `xgb_model.pkl` — Trained XGBoost classifier
- `scaler.pkl` — Fitted StandardScaler for input normalization

All three files are generated at the end of the training notebook using `joblib.dump()` and must be co-located with the `app.py` file at runtime.

9. Limitations & Future Work

9.1 Current Limitations

- **Binary Classification Only:** The system classifies patients as anemic or non-anemic but does not differentiate between anemia subtypes (iron-deficiency, megaloblastic, aplastic, hemolytic, etc.), which require different treatments.
- **Class Imbalance:** The 65:35 class ratio may marginally bias the model toward predicting the majority class. Techniques such as SMOTE or class weighting were not applied in this iteration.
- **No Hyperparameter Tuning:** Both models were trained with default or lightly specified hyperparameters. Grid search or Bayesian optimization could yield improved performance.

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- **Limited Demographic Features:** Age and Gender were excluded from the final feature set due to low correlation. More complex interactions (e.g., age-stratified hemoglobin thresholds) are not captured.
- **No Cross-Validation:** A single train-test split was used. K-fold cross-validation would provide more robust performance estimates and reduce the effect of random variation in the split.
- **Single Dataset:** Generalizability to other patient populations or laboratory instruments has not been validated. External validation on a different cohort is recommended before clinical deployment.

9.2 Future Improvements

- **Anemia Subtype Classification:** Extend to multi-class classification for iron-deficiency, B12 deficiency, folate deficiency, and other subtypes using the same feature set with additional markers.
- **Hyperparameter Optimization:** Implement RandomizedSearchCV or Optuna for automated hyperparameter tuning to potentially improve F1-score and recall.
- **Imbalance Handling:** Apply SMOTE (Synthetic Minority Oversampling Technique) or adjust class weights to improve recall on the minority anemia class.
- **Additional Models:** Evaluate LightGBM, CatBoost, and neural network architectures (e.g., TabNet) as alternative classifiers.
- **SHAP Explainability:** Integrate SHAP (SHapley Additive exPlanations) values into the application to provide patient-level feature contribution explanations for clinician transparency.
- **Longitudinal Analysis:** Incorporate time-series patient data to model anemia progression and treatment response over time.
- **API Deployment:** Expose the prediction model as a REST API using FastAPI or Flask for integration with existing hospital information systems (HIS).

10. Conclusion

This project successfully demonstrates the application of machine learning to clinical anemia prediction using a large-scale hematological dataset. The analysis pipeline — from data exploration through model deployment — provides a complete framework for building automated clinical decision support tools.

The Random Forest classifier achieved the best overall performance with 93.48% accuracy and an F1-Score of 89.77%, supported by a precision of 98.51% that makes it highly reliable in flagging true anemia cases. The XGBoost model offers a complementary trade-off with marginally higher recall, and the ensemble approach in the web application leverages both models for balanced clinical decision support.

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Key clinical insights confirmed by the data analysis include the primacy of Hemoglobin, Hematocrit, and RBC count as anemia predictors, the gender disparity in anemia prevalence, and the diagnostic utility of iron metabolism markers (Serum Iron, Ferritin, TIBC). These findings align with established hematological knowledge, validating the dataset quality and analytical methodology.

The Streamlit application provides an accessible, real-time interface for prediction, making the model practically useful in a screening or educational context. With further validation, hyperparameter optimization, and subtype classification capabilities, this system has strong potential for deployment in clinical laboratory workflows.

Achievement	Value
Dataset Processed	153,728 patient records
Features Engineered	14 hematological parameters
Best Model Accuracy	93.48% (Random Forest)
Best F1-Score	89.77% (Random Forest)
Best Precision	98.51% (Random Forest)
Application Built	Streamlit real-time prediction app